Aravind Boddu

5715819235 email- aboddu2@gmu.edu
LinkedIn - https://www.linkedin.com/in/aravindlohit2398
Github - https://github.com/aravindlohit

EDUCATION

GEORGE MASON UNIVERSITY May 2024
MS - Electrical Engineering, GPA : 3.44/4.0 VA, USA

ANITS (ANDHRA UNIVERSITY) May 2019
Bachelor of technology - Electrical Engineering India

TECHNICAL SKILLS

Tools: Xillinx Vivado and ISE, Modelism Altera, Synopsys VCS, Design Compiler, Prime time, VCS ,IC compiler, LTspice
Scripting Language: Python
HDL/HVL : Verilog, System Verilog, SVA, UVM
Protocols : AXI, AHB, APB, I2C, UART
Relevant Coursework : Computer Architecture, Digital and Analog IC Design, ASIC Design , Digital System Design

EXPERIENCE

1.Design Verification Engineer Dec 2019 – June 2022
Visual Solutions, INDIA
• Comprehensively trained in advanced digital design, RTL coding ,Delved into the Universal Verification Methodology (UVM), gained experince in AMBA Protocols.
• Involved in a UVC Component development for an AHB-compliant peripheral, developing sequence, driver, sequencer, coverage components.
• Developed constrained-random testcases covering burst transfers, arbitration.
• Achieved AXI protocol compliance (valid-ready handshakes, burst length, and response ordering) by incorporating SVA.
• UVM Verification Framework: Developed a reusable UVM testbench for AXI interconnect verification, improving regression efficiency.
• Debugged and resolved 50+ RTL simulation failures, improving the stability.
• Created a System Verilog interface for a legacy bus protocol, facilitating seamless integration of new IP blocks and saving 40 hours of development time.
• Partnered with senior architects to refine micro-architecture specifications, enhancing design clarity for a team of four engineers.

2. Python Data Engineer- Intern (Scripting) Aug 2024 - Dec 2024
Iven Inc, Remote
• Troubleshooting and resolving data-related issues to maintain seamless data flow. Documenting data workflows, processes, and technical specifications.

3. Teaching Assistant - George Mason University Jan 2024 - May 2024
• Evaluated assignments and projects for ECE 101 – Introduction to Electrical and Computer Engineering and Assisted students in designing, building, and testing circuits (filters, RLC oscillators, BJT circuits)

ACADEMIC PROJECTS

1. Behavioral Modeling of a 5-Stage Pipelined CPU (RISC-V) [System Verilog]
• Implemented a pipelined RISC-V processor datapath with instruction fetch, decode, ALU, and write-back stages for R-type and I-type instructions. Integrated basic hazard handling testbench to resolve data dependencies and verified correct execution through waveform simulation.

2. Verification of UART Controller (Open Source) [System Verilog]
• UART core is verified in loopback configuration and analyzed the functionality baud rates, break frame transfers, Constrained randomization is used to generate the relevant stimulus to limit the transactions.

3. Datapath synthesis of the 'XETA cipher' a LWC Core. [Verilog , Vivado]
• Developed RTL Datapath and Controller for XETA Cipher and written fully synthesizable Verilog code Targeting Artix-7 Family,Optimized Datapath to reduce the logic utilization by 30%

4. Performing Power Efficiency Analysis of 'Xoodyak' a LWC Core [DC Compiler,IC Compiler (RTL to GDSII)]
• Evaluated power efficiency of Xoodyak hardware core using two technology nodes, compared metrics by changing constraints for both technologies 90nm and 32nm and constructed GDSII of the core using Synopsys ICC.